Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/03/2004
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Project Title:	Spatial Reorientation of Sensorimotor Balance Control in Altered Gravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77204-6015	**Congressional District:**	18
Comments:	PI moved to University of Houston in September 2008, from NASA JSC.		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	Shuttle		
Flight Assignment:	Pre-/Postflight Only		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Wood, Scott Ph.D. (Universities Space Research Association) Kaufman, Galen Ph.D. (University of Texas Medical Branch) Reschke, Millard Ph.D. (NASA Johnson Space Center) Black, Franklin M.D. (Legacy Health Sysytem)		
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Task Description:	Disruption of balance control is one of the most significant post-flight manifestations of sensorimotor adaptation to microgravity. This investigation examines changes in spatial processing of sensorimotor function following adaptation to microgravity, and explores the feasibility of forcing changes between motor program sets optimized for different gravitoinertial environments. This study includes pre- and post-flight measurements obtained on ten astronaut subjects (preferably first-time fliers) following short-duration Shuttle missions. Each astronaut participates in one familiarization training session (including standard posturography testing), three pre-flight data sessions, and five post-flight data sessions on R+0, 2, 3, 4, and 8 days. Each test session includes posture measurements, and three sessions (L-60, L-30 and R+3) include additional posture measurements immediately after exposure to short-radius centrifugation. In addition, testing on normative subjects support data interpretation and establish ground-based controls. Our first specific aim is to examine adaptive changes in the spatial reference frame used for coding orientation and motion as a function of space flight. By examining the effects of head tilt on balance control, we will test our hypothesis that there is a reorientation of central vestibular processing from a gravitational frame of reference to a head frame of reference as a function of adaptation to microgravity. Our second specific aim is to examine the feasibility of altering the re-adaptation process following space flight by providing discordant canal-otolith-somatosensory stimuli using short-radius pitch centrifugation. This study specifically addresses Neurovestibular Adaptation questions of the Critical Path Roadmap (CPR Risks 15 and 16) related to the physiological basis for disruption of balance control and spatial disorientation following g-transitions. Additional post-flight posturography measures obtained during this study will expand our existing database and increase our ability to predict recovery curves from mathematical models for individuals based on pre-flight postural performance. The incorporation of head tilts during posturography addresses our general hypothesis for the physiological basis for the disruption of balance control following g-transitions, and provides an enhanced technique to determine crew readiness to return to normal activities following return to Earth. The incorporation of short-radius centrifugation during the post-flight period addresses the ability of crewmembers to adapt to multiple gravitoinertial environments and switch between them, e.g., as required for intermittent exposure to artificial gravity. The results of this study will also assess the potential risk for disrupting the re-adaptation process by exposure to discordant sensory stimuli during low levels of centrifugation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research can be used by the NASA research and operational community to evaluate the feasibility of using short arm centrifugation as an in-flight sensorimotor countermeasure. The results are of use to clinical and military researchers interested in adaptive changes in spatial processing to altered gravito-inertial environments, and the fragility of balance control adaptive processes. Findings of the ground-based study on static and dynamic head movements during computerized dynamic posturography may help to improve the sensitivity of the test paradigm in selected clinical populations.
Task Progress:	Normative studies: Several normative studies on non-astronaut participants have been conducted to refine the flight protocol, establish baseline measures and aid in data interpretation. In preparation for the flight study, postural performance was measured in 12 healthy adult humans during static and dynamic (0.14, 0.33, 0.6 Hz) head tilts of ±30° in the pitch and roll planes. While subjects were able to maintain upright stance with static head tilts in lateral or forward directions, postural sway was increased during neck extension. Postural stability was also decreased during dynamic head tilts, and the degree of destabilization varied directly with the frequency of head tilt. The effects of dynamic head tilts were similar for both pitch and roll planes. The results of this study are consistent with our hypothesis that postural stability may be compromised due to a decreased ability to discern the orientation of gravity using vestibular information alone. This study was useful to optimize the astronaut posture protocol by: restricting head tilts to the pitch plane, limiting the frequency of the dynamic tilts to 0.33 Hz, reducing the amplitude of the tilts to 20° and performing statistical power analyses to determine the number of trials required for each condition. These same twelve subjects also participated in a separate centrifuge control study during which the superimposed sinusoidal oscillation varied over this same frequency range. Eleven subjects were also oscillated at 0.14, 0.3, or 0.6 Hz without centrifugation with the interaural axis positioned either 0.5m off-axis and over the axis of rotation. The vertical eye velocity gains were more enhanced during off-axis rotation at lower frequencies, while differences in horizontal vergence in the dark were greater between on- and off-axis conditions at higher frequencies. During visual-vestibular interaction, the vertical gain was systematically enhanced during forward-facing rotation and suppressed during backward-facing rotation across frequencies. The upward bias velocity in the dark was significantly increased during the visual-vestibular interactions, more so in the forward-facing direction. Based on the results of this initial study, we elected to provide two runs for each centrifuge session incorporating oscillation at 0.33 Hz in both forward-facing and backward-facing directions. Pre- and Postflight Measures – Specific Aim 1: Our first specific aim is to examine adaptive changes in the spatial reference frame used for coding orientation and motion as a function of space flight. We predicted the use of a head spatial reference frame will be characterized by differences in the landing day measurements relative to those observed before flight. These include decreased postural stability as evidenced by increased sway amplitudes and decreased equilibrium scores, with greater differences between pre- and post-flight during head tilt conditions. Five of ten astronauts have completed this study, and an ongoing normative study is obtaining repeated postural measures in healthy age, gender and height-matched control subjects using the same timeline scheduling as their astronaut counterparts. Pre-flight scores to date for head erect, forward, backward, and moving at 0.33 Hz in the pitch plane were all in the upper range of the normative performance distributions. Consistent with our previous post-flight studies, performance on landing day was substantially decreased for conditions with absent visual and altered proprioceptive feedback. The most striking result on landing day; however, was on trials requiring active head movements at 0.33 Hz. All subjects fell on SOT-5 trials on landing day with active head movements (9 of 10 trials overall). The recovery curves for these crewmembers for trials with head erect followed similar trajectories to previous astronaut data recorded by our laboratory. The recovery profile for trials with the head movements, in contrast, had a similar recovery time constant but at significantly lower performance levels compared with the head erect trials. This decrease in performance during head tilts is consistent with our hypothesis that the physiological basis for disruption of balance control and spatial disorientation following g-transitions relates to a change in the reference frame used for central vestibular processing. Pre- and Postflight Measures – Specific Aim 2: Our second specific aim is to examine the feasibility of altering the re-adaptation process following space flight by providing discordant canal-otolith-somatosensory stimuli using short-radius pitch axis centrifugation. The centrifuge profile utilized in this study represents a composite of different

	profiles that appeared to disrupt the recovery of one Spacelab astronaut as late as R+5 as described above. Although there were some interesting modulations of vertical and horizontal vergence eye movements during centrifugation, there were no apparent differences in these responses between pre- and post-flight. There were also no consistent differences between motion perception reports from pre- to post-flight. Although the postural performance following centrifugation was also not substantially changed in any of the five crewmembers tested to date in this study, this may reflect behavioral characteristics that contribute to the large inter-individual differences in neurovestibular symptoms. It is interesting to note that pre-flight scores of the five participants were all in the upper range of the normative performance distributions. The risk of centrifugation disrupting the re-adaptation process may be greater for individuals that score in the lower range of pre-flight performance distributions.
Bibliography Type:	Description: (Last Updated: 02/09/2021)
Abstracts for Journals and Proceedings	Paloski WH, Newby NJ, Hwang EY. "Head movement augmented postural control testing on short-duration astronauts." Aerospace Med Assoc 75th Annual Scientific Meeting, Anchorage, Alaska, May 2004. Aviat Space Environ Med 2004 Apr; 175(4 Suppl):B95. , Apr-2004
Abstracts for Journals and Proceedings	Wood SJ, Newby NJ, Kaufman GD, Paloski WH. "Binocular coordination of the vestibulo-ocular reflex during sinusoidally varying short-radius centrifugation." Aerospace Med Assoc 75th Annual Scientific Meeting, Anchorage, Alaska, May 2004. Aviat Space Environ Med 2004 Apr; 175 (4 Suppl):B43. , Apr-2004